

introduction to dynamic

1. List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

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Point	Static Routing	Dynamic Routing	Real-World Example
Configuration	Routes are manually configured by the network administrator.	Routes are automatically learned and updated using routing protocols.	Google Maps: Static routing is like manually choosing one fixed road, while dynamic routing is like Google Maps automatically suggesting a new route.
Route Changes	Does not automatically change if a network path fails.	Automatically finds an alternative route when a path fails.	Swiggy: If one road is closed, dynamic routing is like automatically sending the delivery partner through another route.

Best Use	Best for small and stable networks.	Best for large and frequently changing networks.	Google Maps/Swiggy: Small areas with fixed routes can use static routing, while large networks with changing traffic and routes benefit from dynamic routing.
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2. Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

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Router0
Router#show ip route
Codes: C - connected, S - static, I - IGMP, R - RIP, M - mobile, B - BGP
       O - OSPF, EX - OSPF external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 1 subnets
    C 10.0.0.0 is directly connected, Serial0/0
  20.0.0.0/30 is subnetted, 1 subnets
    R 20.0.0.0 [120/1] via 10.0.0.2, 00:00:16, Serial0/0

Router#show ip protocol
% Invalid input detected at '''' marker.

Router#show ip protocol
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 18 seconds
  Invalid after 180 seconds, hold down 180, flushed after 360
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
  Default version control: send version 2, receive 2
    Interface          Send Recv Triggered RIP  Key-chain
  Serial0/0            22
  Serial0/1            22
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.2         120          00:00:01
  Distance: (default is 120)
  Router#

```

```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

Serial0/0 10.0.0.0 YES manual up up
Serial0/1 20.0.0.1 YES manual up up
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGMP, R - RIP, M - mobile, B - BGP
       O - OSPF, EX - OSPF external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 1 subnets
    C 10.0.0.0 is directly connected, Serial0/0
  20.0.0.0/30 is subnetted, 1 subnets
    C 20.0.0.0 is directly connected, Serial0/1

Router#show ip protocol
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 13 seconds
  Invalid after 180 seconds, hold down 180, flushed after 360
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
  Default version control: send version 2, receive 2
    Interface          Send Recv Triggered RIP  Key-chain
  Serial0/0            22
  Serial0/1            22
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    20.0.0.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.2         120          00:00:01
  Distance: (default is 120)
  Router#

```

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

ROUTER#
VLINEPROTO-5-UPDOWN: Line protocol on Interface Serial10/0, changed state to up
Router#
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, I - IGP
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/30 is subnetted, 1 subnets
     R    10.0.0.0 [120/1] via 10.0.0.1, 00:00:124, Serial10/0
C    20.0.0.0/30 is subnetted, 1 subnets
     C    20.0.0.0 is directly connected, Serial10/0

Router#show ip protocol
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 16 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
    Default version control: send version 2, receive 2
    Interface          Send Recv Triggered RIP Key-chain
    Serial10/0         22
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    20.0.0.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance    Last Update
    20.0.0.1        120         00:00:12
  Distance: (default is 120)
Router#
```

3. Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.

Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

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```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

* Invalids input detected at "" marker.
Router#show ip protocol
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 18 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
    Default version control: send version 2, receive 2
    Interface          Send Recv Triggered RIP Key-chain
    Serial10/0         22
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance    Last Update
    10.0.0.2        120         00:00:01
  Distance: (default is 120)
Router#
Router#
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, I - IGP
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/30 is subnetted, 1 subnets
     C    10.0.0.0 is directly connected, Serial10/0
R    20.0.0.0/30 is subnetted, 1 subnets
     R    20.0.0.0 is possibly down, routing via 10.0.0.2, Serial10/0

Router#
```

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Router2#show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 16 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
    Default version control: send version 2, receive 2
  Interface
    Serial0/0
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    20.0.0.0
  Passive Interface(s):
  Routing Information Sources:
    Gateway         Distance      Last Update
  20.0.0.1          120           00:00:02
  Distance: (default is 120)
Router2#
%LINE-3-UPDOWN: Interface Serial0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
Router2#
Router2#
Router2#
Router2#
Router2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

20.0.0.0/30 is subnetted, 1 subnets
C    20.0.0.0 is directly connected, Serial0/0
R    20.0.0.0/30 is subnetted, 1 subnets
R    20.0.0.0 is possibly down, routing via 10.0.0.2, Serial0/0

Router2#
```

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
C    10.0.0.0 is directly connected, Serial0/0
20.0.0.0/30 is subnetted, 1 subnets
R    20.0.0.0 is possibly down, routing via 10.0.0.2, Serial0/0

Router0#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
C    10.0.0.0 is directly connected, Serial0/0
20.0.0.0/30 is subnetted, 1 subnets
R    20.0.0.0 is possibly down, routing via 10.0.0.2, Serial0/0

Router0#
```

4. Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.



Dynamic routing in apps like Uber or Zomato automatically updates the delivery path when the location or traffic conditions change. Similarly, RIP automatically updates routing tables when network paths change or a route becomes unavailable. Both systems choose and update the best available path automatically instead of requiring manual changes.